Statistics Exam
Master Guide

Multiple Linear Regression & Time Series

How to read questions · Which method to use · Why it works
Common pitfalls · Trick phrasings · Worked decision logic

This guide walks you through every type of question that appears in exams on Multiple Linear Regression (MLR) and Time Series Regression. For each topic you will find: **what the question is really asking**, **which method to choose and why**, **step-by-step decision logic**, **the correct interpretation**, **pitfalls that cost marks**, and **different phrasings that mean the same thing**.

PART 1 — MULTIPLE LINEAR REGRESSION (MLR)

1.1 How to Read an MLR Question

Before touching SPSS, read the question twice and identify four things:

What to identify	How it appears in the question	What you must do
Dependent variable (Y)	"Regress X on Y", "Y as a function of", "explain Y"	Put Y in the Dependent box
Continuous predictors	Age, income, price — no categories mentioned	Enter directly
Categorical predictors	"Region", "type", "group" with labels	Create dummy variables manually
Log transformation	"ln(X)", "log(X)", "logarithm of"	Variable is already logged OR use Transform → Compute
Interaction term	"effect of X depends on Z", "moderating effect"	Create X×Z product variable

1.2 Dummy Variables

A dummy variable (also called indicator variable or binary variable) converts a categorical variable with k categories into k–1 binary (0/1) variables. The excluded category is the **reference group** — all interpretations are made relative to it.

Rule: k categories → create k–1 dummies

Categories	Dummies to create	Reference (omitted)
North, South, Center, East (4)	D_North, D_South, D_East (3)	Center
Male, Female (2)	D_Female (1)	Male
Low, Medium, High (3)	D_Medium, D_High (2)	Low

In SPSS: Transform → Compute Variable

For each dummy, type an expression that equals 1 for that group:

```
D_North = (Region = 1) [if North is coded as 1]

D_South = (Region = 2)

D_East = (Region = 4) [skip 3 = Center → reference]
```

How to write the theoretical model:

```
Y = β■ + β■X■ + β■D_North + β■D_South + β■D_East + ε


Interpretation of β■ (D_North):
```

```
On average, North regions have β■ units MORE Y than Center regions,

holding all other variables constant.
```

■ Common Pitfalls

- Never create k dummies — always k–1. Including all categories causes perfect multicollinearity (the dummy trap). SPSS will drop one automatically, but you must know which one is the reference.
- The question may say "use Center as reference" — that just means do NOT create a dummy for Center.
- Different phrasings for the same thing: "baseline group", "comparison group", "omitted category", "reference category" — all mean the same thing.

1.3 Log Transformations and Interpretation

Model type	Equation	Interpretation of β
Linear (standard)	$Y = \alpha + \beta X + \varepsilon$	1-unit rise in X → β unit change in Y
Linear-log	$Y = \alpha + \beta \cdot \ln(X) + \varepsilon$	1% rise in X → $\beta/100$ unit change in Y
Log-linear	$\ln(Y) = \alpha + \beta X + \varepsilon$	1-unit rise in X → $\beta \times 100$ % change in Y
Log-log	$\ln(Y) = \alpha + \beta \cdot \ln(X) + \varepsilon$	1% rise in X → β% change in Y (elasticity)

Linear-Log Worked Example

- The most common exam type is Linear-log: $Y = \alpha + \beta \cdot \ln(X) + \varepsilon$.
- Trick: the question gives $\beta = -16$ and X is ln(NOx). Answer: 1% increase in NOx → $-16/100 = -0.16$ unit change in Y (e.g., –$160 if Y is in thousands).
- Key phrase to watch for: 'ln(X)' or 'log(X)' in the variable name in the output.

1.4 Interpreting Coefficients — The Full Decision Tree

When the question says 'interpret the coefficient of X', follow this tree:

Is X logged?	Is Y logged?	Interpretation
No	No	1 extra unit of X → β extra units of Y (ceteris paribus)
Yes (ln X)	No	1% extra in X → $\beta/100$ extra units of Y
No	Yes (ln Y)	1 extra unit of X → $\beta \times 100$ % change in Y
Yes (ln X)	Yes (ln Y)	1% extra in X → β% change in Y
Dummy (0/1)	No	Being in group D → β extra units of Y vs. reference

1.5 Testing Individual Coefficient Significance

Every coefficient in the SPSS Coefficients table has a t-statistic and a p-value (Sig. column). Use the p-value directly:

```
H0: β = 0 (variable has no effect)

H1: β ≠ 0 (variable does have an effect)


Decision: If Sig. < 0.05 → Reject H0 → significant at 5%

If Sig. ≥ 0.05 → Fail to reject H0 → not significant
```

■ Common Pitfalls

• "Significant" does NOT mean "important" or "large". It means we can statistically distinguish the effect from zero.

• Different phrasings: "Is X relevant?", "Does X matter?", "Test whether X has an effect", "Is β significantly different from zero?" — all ask for the t-test/p-value.

1.6 Joint F-test (Testing a Group of Variables Together)

Use a joint F-test when the question asks whether a GROUP of variables (e.g., all region dummies together) is significant, not just one.

When do you know to use a joint F-test?

Trigger phrases for joint F-test

• "Are the region dummies jointly significant?"

• "Do the seasonal effects together improve the model?"

• "Test whether including dummies D1, D2, D3 adds explanatory power."

• "Is the set of interaction terms significant?"

• Any question involving MORE THAN ONE variable being tested simultaneously.

How to run it in SPSS:

Step	What you do	Why
1	Run the UNRESTRICTED model — all variables included	Get SSE_u (or R^2_u)
2	Run the RESTRICTED model — remove the variables being tested	Get SSE_r (or R^2_r)
3	Count J = number of variables you removed	J = numerator df
4	Get n and k from unrestricted model	n = observations, k = predictors
5	Compute F using the formula	Compare to p-value

```
Option A (using SSE from ANOVA tables):

F = [(SSE_r − SSE_u) / J] / [SSE_u / (n − k − 1)]
```

```
Option B (using R² from Model Summary):
F = [(R²_u − R²_r) / J] / [(1 − R²_u) / (n − k − 1)]

Decision: Use the p-value from SPSS for the overall F in the ANOVA table.
If you compute F manually: compare to F-distribution critical value.
Shortcut: If both ANOVA Sig. values are available, compare them.
```

■ Common Pitfalls

- J is the number of RESTRICTIONS (variables removed), not the total number of variables.
- The restricted model has FEWER variables. Its SSE will always be LARGER (worse fit).
- n–k–1 uses k from the UNRESTRICTED model.
- "Test whether dummies are jointly significant" = run restricted model WITHOUT those dummies.

1.7 Residual Plot Analysis

A residual plot (residuals on Y-axis, fitted values or X on X-axis) is used to detect three violations of OLS assumptions:

Problem	What you see in the plot	Assumption violated
Heteroskedasticity	Spread of residuals grows or shrinks — fan shape (wider on one side)	A2: Var(ε) = σ^2 (constant variance)
Non-linearity	Residuals curve upward or downward — systematic pattern	A1: Model is correctly specified (linearity)
Outliers	One or a few points far above/below the rest — isolated large residuals	Distort estimates — check and report

What a good vs. bad plot looks like

- A GOOD residual plot: random cloud of points centered around zero. No pattern, no fan, no curve.
- A BAD plot: any systematic shape — V-shape, cone, curve, or obvious outliers.
- You do NOT need to quantify — describe what you SEE and name the violation.

■ Common Pitfalls

- Do not say 'the residuals are not normal' from a residual vs. fitted plot — normality requires a histogram or Q-Q plot.
- Heteroskedasticity in residuals → standard errors are wrong → t-tests and p-values are unreliable. This is why we run the White test.

1.8 The White Test for Heteroskedasticity

The White test formally tests whether the variance of the error is constant (homoskedastic) or varies (heteroskedastic).

```
H0: homoskedasticity (Var(ε) = σ² — constant)

H1: heteroskedasticity (variance is not constant)


Step 1: Run original model. Save residuals ê.

Step 2: Compute ê² (squared residuals) via Transform → Compute.

Step 3: Run auxiliary regression:

ê² = α■ + (all original X's) + (X² for each continuous X) + (Xi×Xj for each pair) + u

Note: do NOT square dummies. Do NOT create dummy × dummy terms.

Step 4: W = n × R²_aux ~ χ²(df)

df = number of regressors in auxiliary regression (excluding constant)

Step 5: If W > critical value (or Sig. < 0.05) → Reject H0 → heteroskedasticity present
```

■ Common Pitfalls

- The df for the White test is NOT 1 — it equals the number of terms in the auxiliary regression. With 2 continuous X's: original X1, X2 + squared $X1^2$, $X2^2$ + cross X1×X2 = 5 df.
- Dummies are never squared ($D^2 = D$ for a 0/1 variable) and dummy×dummy products are excluded from the auxiliary regression.
- Always state df in your answer: $W \sim \chi^2(df)$. Different questions have different df.

1.9 Omitted Variable Bias

If a relevant variable Z is left out of the model, the estimated coefficient on the included variable X will be biased. The direction of the bias is:

```
sign(bias on X) = sign(β_Z) × sign(r(X, Z))


Where: β_Z = the true effect of the omitted variable Z on Y

r(X, Z) = correlation between included X and omitted Z


If bias > 0: estimate is TOO HIGH (upward bias)

If bias < 0: estimate is TOO LOW (downward bias)
```

β_Z (effect of Z on Y)	r(X, Z) (correlation)	Bias on β_X	Estimate is...
Positive (+)	Positive (+)	Positive → upward bias	Too high
Positive (+)	Negative (–)	Negative → downward bias	Too low
Negative (–)	Positive (+)	Negative → downward bias	Too low

Negative (–)	Negative (–)	Positive → upward bias	Too high

Worked Example

• Worked example: Estimating effect of NOx pollution on housing values, omitting industrial zone variable (Indus).

• β_Indus < 0 (more industry → lower house values). r(lnNOx, Indus) > 0 (NOx and industry are positively correlated).

• Bias = (–) × (+) = negative → the estimate of β_lnNOx is TOO LOW (more negative than truth).

■ Common Pitfalls

• "Biased downward" means the estimate is too small (too negative for a negative coefficient).

• The question might not give you the correlation directly — reason from common sense: do areas with high NOx also tend to have more industry? If yes, correlation is positive.

• "Omitted variable bias" and "specification error" mean the same thing.

1.10 VIF and Multicollinearity

Multicollinearity occurs when two or more predictors are highly correlated with each other. This inflates standard errors and makes individual t-tests unreliable even if the model fits well.

```
VIF_j = 1 / (1 − R²_j)

Tolerance_j = 1 − R²_j = 1 / VIF_j


Where R²_j = R² from regressing X_j on all other X's (auxiliary regression)


Rule of thumb: VIF > 5 (or 10) → serious multicollinearity

Tolerance < 0.2 (or 0.1) → serious multicollinearity
```

VIF value	Tolerance	Interpretation
1.0	1.00	No multicollinearity — X is uncorrelated with other X's
1 – 5	0.20 – 1.00	Acceptable — moderate correlation, no problem
5 – 10	0.10 – 0.20	Concerning — high multicollinearity, check estimates
> 10	< 0.10	Severe — estimates are unreliable

• In SPSS: Analyze → Regression → Linear → Statistics → tick Collinearity diagnostics.

• VIF and Tolerance appear in the Coefficients table.

• If asked to compute VIF manually: run the auxiliary regression of X_j on all other X's, read R^2, then VIF = $1/(1 - R^2)$.

1.11 Quick Reference — Question Phrasings and Methods

If the question says...	It means...	Method
"Interpret the coefficient of ln(X)"	Effect of 1% change in X	β/100 units change in Y
"Test if Region is significant"	All region dummies jointly	Joint F-test
"Is β■ significantly different from zero?"	Individual significance	p-value from Coefficients table
"Test for heteroskedasticity"	Unequal error variance	White test ($W = n \times R^2\_aux$)
"What is the bias if Indus is omitted?"	Direction of bias	sign(β_omitted) × sign(r(X, omitted))
"What is the reference category?"	Omitted dummy group	State which group has no dummy variable
"VIF is 2.5 — what does this mean?"	Multicollinearity assessment	Compare to threshold of 5
"Write the theoretical model"	Symbolic equation with all variables	Write Y = β■ + β■X■ + β■D■ + ...

PART 2 — TIME SERIES REGRESSION

2.1 The Three-Step Framework — Always in This Order

The Three-Step Framework

- STEP 1 — STATIONARITY: Test whether each series is stationary (Dickey-Fuller test). Non-stationary series must be differenced before use.
- STEP 2 — FIT THE MODEL: Choose static, DL(q), or AR(1) model depending on what the question specifies. For DL(q), use top-down lag selection.
- STEP 3 — CHECK AUTOCORRELATION: Run the LMSC test on the residuals of the fitted model.

■ Common Pitfalls

- You CANNOT skip Step 1. Using non-stationary series in regression produces spurious results — high R^2 and significant t-statistics that mean nothing.
- Steps must go in order: stationarity → fit → check. Never fit first.

2.2 Step 1 — Stationarity and the Dickey-Fuller Test

A stationary series has a constant mean and variance over time. A non-stationary series drifts or wanders — its mean changes. Using non-stationary series causes spurious regression.

How to choose DF Version 1 or Version 2:

DF Version	Regression run	When to use
Version 1 — no trend	ΔY_t = α■ + α■Y_{t-1} + ε_t	Time plot shows NO visible upward or downward trend
Version 2 — with trend	ΔY_t = α■ + λt + α■Y_{t-1} + ε_t	Time plot shows clear trend, OR when unsure (safer default)

```
H0: α■ = 0 (unit root — series is NON-STATIONARY)

H1: α■ < 0 (series IS stationary)


Critical values (5% level):

Version 1 (no trend): −2.86

Version 2 (with trend): −3.41


Decision: If t-statistic on Y_{t-1} is MORE NEGATIVE than the critical value → Reject H0
→ stationary

If t-statistic is LESS NEGATIVE (closer to zero) → Fail to reject H0 → unit root
```

```
Example: t = −1.64, critical value = −3.41

−1.64 > −3.41 → fail to reject H0 → NON-STATIONARY


Example: t = −4.20, critical value = −3.41

−4.20 < −3.41 → reject H0 → STATIONARY
```

What to do after the DF test:

DF result on levels	Action	Then what?
Stationary (reject H0)	Use series in levels	Proceed to fit model
Non-stationary (fail to reject H0)	Take first differences: $\Delta Y = Y_t - Y_{t-1}$	Run DF test again on ΔY
ΔY is stationary (reject H0)	Use first differences in model	Series is I(1) — proceed
ΔY still non-stationary	Rare — take second differences	Series may be I(2)

■ Common Pitfalls

• The DF test has a ONE-SIDED alternative: H1 is $\alpha_{\blacksquare} < 0$, not $\alpha_{\blacksquare} \neq 0$. You only reject when the t-statistic is sufficiently NEGATIVE.

• NEVER use the standard t-table (±1.96) for the DF test. The DF distribution is non-standard. Always use −2.86 (V1) or −3.41 (V2) at 5%.

• In SPSS, the p-value shown for the LAG variable uses the wrong distribution — ignore it. Compare only the t-statistic to the special critical value.

• Rejecting H0 in the DF test means the series IS stationary — opposite of most tests where rejecting H0 is the 'bad news'.

2.3 Step 2 — Which Model to Fit?

Model	Equation	When to use
Static	$Y_t = \alpha + \beta X_t + \varepsilon_t$	Question specifies it, or no significant lags found in DL
DL(q) — Distributed Lag	$Y_t = \alpha + \beta_{\blacksquare} X_t + \beta_{\blacksquare} X_{t-1} + \ldots + \beta_q X_{t-q} + \varepsilon_t$	Question asks for DL model; effects of X are delayed over time
AR(1) — Auto regressive	$Y_t = \alpha + \beta X_t + \gamma Y_{t-1} + \varepsilon_t$	Question specifies AR(1); or strong autocorrelation in DL residuals

2.4 DL(q) — Top-Down Lag Selection

The top-down procedure finds the optimal number of lags q by starting high and removing insignificant lags one at a time from the top.

```
Start: Fit DL(q_max). In SPSS, create lag variables first:
```

```
LAG1_X = LAG(X, 1) LAG2_X = LAG(X, 2) etc.

Top-down rule:

1. Look at the p-value of the HIGHEST lag (e.g., LAGq_X).

2. If p > 0.05: not significant → drop it → fit DL(q-1).

3. If p ≤ 0.05: significant → STOP. Optimal q is the current model.

4. Repeat until the highest remaining lag is significant.

IMPORTANT: Never drop an intermediate lag.

If lag 2 is significant but lag 1 is not → keep BOTH.

You cannot have lag 2 without lag 1 in the model.
```

Total Multiplier for DL(q):

```
Total multiplier = β■ + β■ + β■ + … + βq

Interpretation: If X permanently increases by 1 unit and stays there forever,

Y will eventually change by this total amount in the long run.

Immediate effect (impact multiplier) = β■ only.
```

■ Common Pitfalls

- q_max is usually given in the question. If not, a common default is 3 or 4.
- Each lag costs you one observation. DL(3) with n=100 gives 97 usable observations.
- "Optimal lag length" and "best-fitting DL model" and "chosen q" all mean the same thing.
- Do not confuse total multiplier (long-run effect) with β■ (immediate effect). The question may specifically ask for one or the other.

2.5 AR(1) — Autoregressive Model

```
Model: Y_t = α + βX_t + γY_{t-1} + ε_t

Coefficients:

β = immediate effect of a 1-unit rise in X_t on Y_t

γ = persistence — how much of last period's Y carries into this period

Requires |γ| < 1 for the model to be valid (stationary errors)
```

```
Total multiplier = β / (1 − γ)

Interpretation: permanent 1-unit rise in X eventually changes Y by β/(1−γ)


Example: β = −0.031, γ = 0.52

Total multiplier = −0.031 / (1 − 0.52) = −0.031 / 0.48 = −0.065
```

■ Common Pitfalls

• In AR(1) with autocorrelation, estimates are BIASED (worse than inefficient). In DL with autocorrelation, estimates are only INEFFICIENT (standard errors wrong). This is why the LMSC test matters after fitting AR(1).

• γ must be strictly less than 1 in absolute value. If $\gamma \geq 1$ the total multiplier formula breaks down (series is non-stationary).

• "Persistence coefficient", "AR coefficient on lagged Y", "coefficient on Y_{t-1}" — all refer to γ.

2.6 Step 3 — The LMSC Autocorrelation Test

Autocorrelation means the error in one period is correlated with the error in the previous period. It violates Assumption A3 and makes standard errors wrong.

```
H0: ρ = 0 (no autocorrelation — A3 satisfied)

H1: ρ ≠ 0 (autocorrelation present — A3 violated)


Step 1: Fit original model. Save Unstandardized Residuals (RES_1).

Step 2: Create LAG_RES1 = LAG(RES_1, 1) via Transform → Compute.

Step 3: Run auxiliary regression:

RES_1 = α■ + α■X■_t + … + α■X■_t + α_{k+1}·LAG_RES1 + u_t

(Include ALL original predictors PLUS the lagged residual)

Step 4: LM = n_aux × R²_aux where n_aux = n − 1 (one obs lost to lag)

Step 5: LM ~ χ²(1). Critical value at 5% = 3.841

If LM > 3.841 (or p < 0.05): Reject H0 → autocorrelation present

If LM ≤ 3.841 (or p ≥ 0.05): Fail to reject H0 → A3 plausibly satisfied
```

How to get n_aux:

• From the ANOVA table of the auxiliary regression: n_aux = Total df + 1.

• Or: n_aux = n – 1 (you lose one observation creating the lag).

• For a DL(q) model: n_aux = n – q – 1 (you lose q obs from lags plus 1 from residual lag).

■ Common Pitfalls

• The auxiliary regression MUST include all original X predictors, not just the lagged residual.

• For a DL(q) model, the auxiliary regression includes: all original X's AND all lagged X's used in the DL model, plus LAG_RES.

• "Test for autocorrelation", "check A3", "Breusch-Godfrey test", "LMSC test" — all refer to the same procedure.

• DW (Durbin-Watson) is NOT valid when Y_{t-1} appears in the model (AR models). Always use LMSC.

2.7 Critical Pitfall Grid — Opposites That Confuse Everyone

Test	H0 means...	Reject H0 means...	Fail to reject means...
Dickey-Fuller	Series HAS a unit root (NON-STATIONARY)	Series IS stationary ✓	Series is non-stationary — difference it
LMSC	NO autocorrelation (A3 satisfied)	Autocorrelation IS present — fix model	A3 plausibly satisfied ✓
Individual t-test	$\beta = 0$ (variable has NO effect)	Variable DOES have a significant effect	Cannot conclude variable matters
Joint F-test	All tested $\beta = 0$ (group has NO effect)	Group of variables jointly significant	Cannot conclude group matters
White test	Homoskedasticity (A2 satisfied)	Heteroskedasticity present — A2 violated	A2 plausibly satisfied ✓

2.8 Deterministic vs. Stochastic Trends — Different Fix!

Two types of non-stationarity exist and they require DIFFERENT fixes. Applying the wrong fix makes things worse.

Feature	Deterministic trend	Stochastic trend (unit root)
Cause	λt term — mean grows linearly	$\rho = 1$ — shocks are permanent
Pattern in plot	Smooth, predictable upward drift	Erratic, wanders unpredictably
Variance	Constant	Grows over time
After a shock	Series returns to trend line	Series never recovers
Correct fix	Add t (time variable) as regressor	Take first differences ΔY
Wrong fix consequence	First differencing → artificial negative autocorrelation	Detrending does not remove random walk

Decision rule

• How to tell them apart: look at the DF test result.

• If DF rejects H0 (stationary) but the plot shows a trend → deterministic trend ($|\rho| < 1$, $\lambda \neq 0$) → add t as regressor.

• If DF fails to reject H0 (unit root) → stochastic trend ($\rho = 1$) → take differences.

2.9 Time Series — Question Phrasings Reference

If the question says...	It means...	Method
"Is the series stationary?"	Does it have a unit root?	Dickey-Fuller test
"Test for a unit root"	Is $\rho = 1$?	Dickey-Fuller test
"Should we use levels or differences?"	What stationarity property does it have?	Run DF, then decide
"Determine optimal lag length"	What is the best q for DL(q)?	Top-down procedure
"What is the long-run effect?"	Total multiplier	DL: $\Sigma\beta k$ | AR: $\beta/(1-\gamma)$
"What is the immediate effect?"	Impact multiplier	β■ (the coefficient on current X only)
"Test for autocorrelation"	Is A3 violated?	LMSC test ($LM = n \times R^2$_aux)
"Is the model well-specified?"	Check all assumptions	LMSC + residual plot
"Series is I(1)"	Non-stationary in levels, stationary in differences	Use first differences in model
"Spurious regression risk"	Non-stationary series used in levels	Run DF first, then decide

2.10 SPSS Step-by-Step Workflow — Full Exam Procedure

Step	SPSS action	What to record
1a. Plot series	Graphs → Chart Builder → Line → drag variable and Time to axes	Does it trend? Random walk?
1b. DF on levels	Create LAG_Y = LAG(Y,1) and DIFF_Y = Y – LAG_Y via Compute. Regression: DV=DIFF_Y, IV=LAG_Y (+ time variable for V2)	t-stat on LAG_Y. Compare to –2.86/–3.41
1c. If unit root: DF on differences	Create LAG_DIFFY = LAG(DIFF_Y,1) and DIFF2_Y = DIFF_Y – LAG_DIFFY. Regression: DV=DIFF2_Y, IV=LAG_DIFFY (+ time for V2)	t-stat on LAG_DIFFY. Compare to –2.86/–3.41
2a. Create lag variables	Transform → Compute: LAG1_X = LAG(X,1), LAG2_X = LAG(X,2) etc.	Note observations lost
2b. Fit model	Regression with appropriate variables (static/DL/AR)	All coefficients, R^2, n
3a. Save residuals	In regression dialog: Save → Unstandardized Residuals	Variable RES_1 created
3b. Create lagged residual	Transform → Compute: LAG_RES1 = LAG(RES_1, 1)	Variable LAG_RES1 created
3c. LMSC auxiliary regression	Regression: DV=RES_1, IV=all original X's + LAG_RES1	R^2 and n from Model Summary + ANOVA
3d. Compute LM	LM = $n_{aux} \times R^2_{aux}$. Compare to 3.841	State conclusion about A3

PART 3 — MASTER PITFALL & TRICK PHRASING LIST

3.1 The Top 15 Mistakes (and How to Avoid Them)

1. Using p-value for DF test	SPSS's p-value for the LAG variable in the DF regression uses the wrong distribution. Always compare the t-statistic directly to –2.86 (V1) or –3.41 (V2).
2. Rejecting H0 in DF = non-stationary	It is the OPPOSITE: rejecting H0 in the DF test means the series IS stationary. H0 is the unit root (non-stationarity).
3. Creating k dummies instead of k–1	For k categories, create exactly k–1 dummies. Including all k causes perfect multicollinearity (dummy trap). Always omit one reference category.
4. Using standard t-table for DF critical values	The DF t-statistic does not follow a standard t-distribution. Standard values like ±1.96 or ±2.58 do not apply. Use –2.86 or –3.41.
5. Forgetting to include all X's in LMSC auxiliary regression	The LMSC auxiliary regression regresses residuals on ALL original X's PLUS the lagged residual. Missing any X gives wrong R^2 and wrong LM statistic.
6. Total multiplier = β■ only	For DL(q): total multiplier = β■ + β■ + … + βq (sum of ALL lag coefficients). For AR(1): total multiplier = β/(1–γ). β■ alone is only the immediate effect.
7. Using DW test for AR(1) models	The Durbin-Watson statistic is INVALID when Y_{t-1} appears as a regressor. Always use the LMSC test.
8. Differencing when trend is deterministic	Only difference when the DF test confirms a unit root ($\rho = 1$). If the series has a deterministic trend ($\lvert\rho\rvert < 1$, $\lambda \neq 0$), add t as a regressor instead.
9. Dropping intermediate insignificant lags in DL	In top-down selection, only drop the HIGHEST lag if insignificant. Never drop lag 1 while keeping lag 2 — the lag structure must be contiguous.
10. Squaring dummies in the White test	$D^2 = D$ for a 0/1 variable, so squaring dummies adds no information. Only square continuous variables. Also exclude dummy × dummy cross-products.
11. Confusing 'significant' with 'large'	A coefficient can be statistically significant ($p < 0.05$) but economically negligible in size. Always comment on both significance AND magnitude.
12. Wrong sign for omitted variable bias	Bias = sign(β_omitted) × sign(r(X_included, X_omitted)). Draw a sign table if needed — two negatives give a positive bias.
13. Ignoring the reference category when interpreting dummies	The coefficient on D_North means 'compared to the reference group'. Always state what the reference category is in your interpretation.

14. Using levels when both series are I(1)	If both Y and X are non-stationary I(1) series, always use first differences (unless cointegration is established — rare in introductory courses).
15. n in LM formula is n_aux not original n	The auxiliary regression loses one observation due to the lagged residual. Use $n_{aux} = n - 1$ (or read from the ANOVA table: n_{aux} = Total df + 1).

3.2 Trick Phrasings — Same Question, Different Words

What the question writes	What it actually means
"Is X relevant to explaining Y?"	"Is β_X significantly different from zero?" → p-value test
"Does adding Region improve the model?"	"Are the region dummies jointly significant?" → Joint F-test
"Baseline group" / "comparison group"	"Reference category" — the omitted dummy group
"Disturbance term carries memory"	"There is autocorrelation" → LMSC test
"Series wanders without returning"	"Series has a unit root" → DF test, then difference
"Spurious correlation risk"	"Series is non-stationary" → run DF before modelling
"Long-run equilibrium effect"	"Total multiplier" → $\Sigma\beta k$ or $\beta/(1-\gamma)$
"Impact effect of a price shock"	"Immediate effect" → β■ only
"Error variance is not constant"	"Heteroskedasticity" → White test
"Predictors are too correlated"	"Multicollinearity" → VIF and Tolerance
"Coefficient on lagged Y is 0.7"	"$\gamma = 0.7$ in AR(1)" → total multiplier = $\beta/(1-0.7)$
"Integrated of order one"	"I(1) — unit root in levels, stationary in differences"
"Misspecification bias"	"Omitted variable bias" → $\text{sign}(\beta_Z) \times \text{sign}(r(X,Z))$
"First-order autocorrelation"	"AR(1) error: $\varepsilon_t = \rho\varepsilon_{t-1} + u_t$" → LMSC test

QUICK DECISION GUIDE — Which Method for Which Question?

MLR Questions

Question type	Method	Key formula / output
Interpret coefficient of ln(X)	Linear-log rule	$\beta/100$ units change in Y per 1% rise in X
Test one coefficient	Individual t-test	p-value (Sig.) in Coefficients table
Test a group of variables	Joint F-test	$F = [(SSE_r - SSE_u)/J] / [SSE_u/(n-k-1)]$
Check equal variance	White test	$W = n \times R^2_{aux} \sim \chi^2(df)$

Direction of missing-variable distortion	Omitted variable bias	$\text{sign}(\beta_Z) \times \text{sign}(r(X,Z))$
How correlated are predictors?	VIF / Tolerance	$VIF = 1/(1-R^2_{aux})$, threshold = 5
Write equation with categories	Dummy variables	k–1 dummies, state reference

Time Series Questions

Question type	Method	Key output / decision
Is series stationary?	Dickey-Fuller test	$t < -2.86$(V1) or -3.41(V2) → stationary
Series drifts with time	DF test → unit root?	Yes → difference. No → add t
Choose number of lags	Top-down DL selection	Drop highest lag while $p > 0.05$
Long-run effect of X on Y	Total multiplier	DL: $\Sigma\beta k$ | AR: $\beta/(1-\gamma)$
Immediate effect of X on Y	Impact multiplier	β■ — coefficient on current X only
Check for autocorrelation	LMSC test	$LM = n \times R^2_{aux}$. Compare to 3.841
Model has Y_{t-1} as regressor	AR(1)	Do NOT use DW. Use LMSC.

✓ Exam Tips

- Always state hypotheses (H0 and H1) before every test — even when they seem obvious.
- Always write a conclusion sentence after every test: 'Since p = X < 0.05, we reject H0...'
- For DF: compare t-statistic (not p-value) to special critical value.
- For LMSC and White test: use p-value directly (standard chi-squared distribution).
- For individual t-tests and F-tests in regression: use p-value from SPSS output.
- When in doubt about stationarity: use DF Version 2 (with trend) — it is the safer default.